HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed **MES Syncade** master recipes and configurations, adhering to corporate design standards for high product quality and lean manufacturing initiatives.
- Executed testing and **computer system validation** for MES recipes and associated configurations, ensuring compliance with regulatory and quality objectives.
- Managed **MES system administration** for third-party interfaces, including SAP ERP, LIMS, and DCS, facilitating seamless data exchange across manufacturing systems.
- Built **MES equipment**, material management, and reporting modules within Syncade, enhancing operational visibility and control for the manufacturing floor.
- Authored and maintained **design documents** and validation deliverables as part of the MES application lifecycle, supporting project documentation requirements.
- Collaborated with cross-functional teams, including operations and business analysts, to gather **requirements** for MES system enhancements and troubleshoot production issues.
- Containerized core Java Spring Boot services using **Docker** and deployed them to an **AWS** EKS cluster, managing configuration via Helm charts for repeatable deployments.

SKILLS

MES Platforms – Syncade, Emerson MES, PASX

Programming Languages – Visual Basic, Java, Python, SQL

Databases – Microsoft SQL Server, PostgreSQL, Oracle, MySQL

Cloud & Containerization – AWS, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes

DevOps & CI/CD – Jenkins, Git, Jira, Confluence

Scripting & Automation – Python, Shell Scripting

Manufacturing Systems – SAP ERP, LIMS, CMMS, DCS

PROJECTS

MES Recipe Development for Pharmaceutical Batch Processing

- Designed and implemented complex master recipes in **MES Syncade** for a critical pharmaceutical batch manufacturing process, covering material management and equipment sequencing.
- Developed custom reports and dashboards within the MES platform to track batch progress, identify deviations, and provide operational insights to stakeholders.
- Collaborated with process engineers and quality assurance teams to validate recipe logic and ensure alignment with Good Manufacturing Practices (GMP) and facility goals.

Java Spring Boot Service for Manufacturing Data Integration

- Engineered a **Java Spring Boot** REST API to integrate with an external **LIMS** system, enabling automated data ingestion for quality control reporting.
- Implemented data validation and error handling logic using **Hibernate** and PostgreSQL to ensure data integrity between the MES and LIMS platforms, processing ~1,500 records daily.
- Containerized the application with **Docker** and deployed it to an **AWS** EC2 instance, establishing CI/CD pipelines using Jenkins for automated builds and deployments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)